Location-based solutions in the Experience centre

Dan Witzner Hansen[1*], Alexandre Alapetite[2], Nanna Holdgaard[1], Celia Simonsen[1], René Larsen Vilsholm[3]

1. IT University of Copenhagen. Rued Langgaards Vej 7. DK-2300 Copenhagen S. Denmark. {witzner, nholdgaard, celiasi}@itu.dk
2. Technical University of Denmark. Management Engineering. Risø Campus, Frederiksborgvej 399, P.O. Box 49, DK-4000 Roskilde, Denmark. alexandre@alapetite.net
3. NaturBornholm. Grønningen 30. DK-3720 Aakirkeby. Denmark. Rene.Larsen@naturbornholm.dk

* Corresponding author. Telephone: +45.72.18.50.88 . Fax: +45.72.18.50.01

Abstract

In this paper we present a prototype system for location-based guiding. A user survey has been conducted and the observations are used to support design choices. The prototype allows for both indoor and outdoor navigation at and in the vicinity of the experience centre NaturBornholm[1] in Denmark using a combination of Bluetooth, GPS and QR-codes. Bluetooth and GPS are used for location-based information and QR-codes are used to convey user preferences.

Keywords

Museum, location-based services, mobile, barcode, GPS, Bluetooth

1 Introduction

Museums, experience centres and tourist organisations continuously try to improve their exhibitions or attractions through a higher degree of user participation by looking at new and interesting ways to stimulate the user experience and increase dissemination.

[1] http://naturbornholm.dk

Location-based services are obvious choices for disseminating knowledge and information in outdoor environments and bridge the gap between indoor exhibitions and outdoor attractions.

Several examples of GPS-based systems are currently being experimented and new ones are being built. Other electronic information systems built upon Bluetooth technology is being investigated[2] for indoor use (e.g. transfer of multimedia contents and location information). Each of these technologies has their strength and weaknesses in terms of applicability to various conditions. For instance, GPS is only appropriate outdoor.

Internationally, a related example is the Ubiquitous Museum[3] where the concept is about displaying posters with QR-codes around the district of Marunouchi (Tokyo, Japan) containing online stories and tourist information. There is also an experience in United Kingdom aiming to provide visitors location-based information about the surroundings of various seaside locations via 2D barcodes (O'Hara & Kindberg, 2007).

In some cities in Denmark, visitors can be guided by using QR-codes and downloading audio files to their mobile phones[4]. More generally, the concept of using barcodes for carrying e.g. information, identification, payment and preferences from a computer to another through the user's mobile phone, is used for so-called "mobile-ticketing", and has

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[2] http://www.kulturarv.dk/tjenester/kulturnet/stoettede_projekter/2008/

[3] http://map.elp.or.jp/umm.html

[4] http://www.visithorsens.dk

several large scale uses, such as the recent 2D-barcode adoption by IATA for check-in[5] with mobile phones at airports (International Air Transport Association, October 2007).

The presented project investigates the possibility to combine technologies that allow visitors of a museum to prioritise and customise the information by creating a profile before the visit while allowing this visitor to bring selected information from inside the museum to the surrounding area outside.

2 Preliminary User Survey

In this section we present the outcome of surveys and interviews in spring 2008 regarding expectations and pre-understandings of museums and the museum visit. From an internet questionnaire we received 138 answers from actual and potential museum visitors in the age 14 – 71 across Denmark. Additionally, a focus group interview involved seven girls from 10th form in Roskilde Tiendeklassecenter.

On one hand our respondents find museums to be uninteresting, rigid, and too elitist (Figure 1), but on the other hand they expect to have a learning experience, have a nice time with friends and family, and use the museum visit in their impression management (Goffman, 1959) (Figure 2).

[5] http://www.iata.org/pressroom/pr/2007-11-10-01.htm

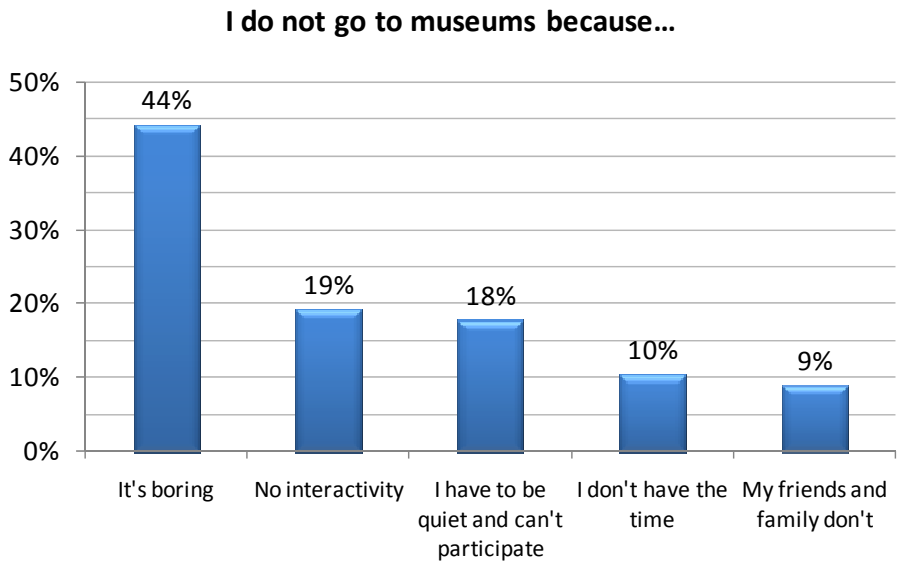


Figure 1: Reasons for not visiting museums, N = 138. The respondents could choose three answers.

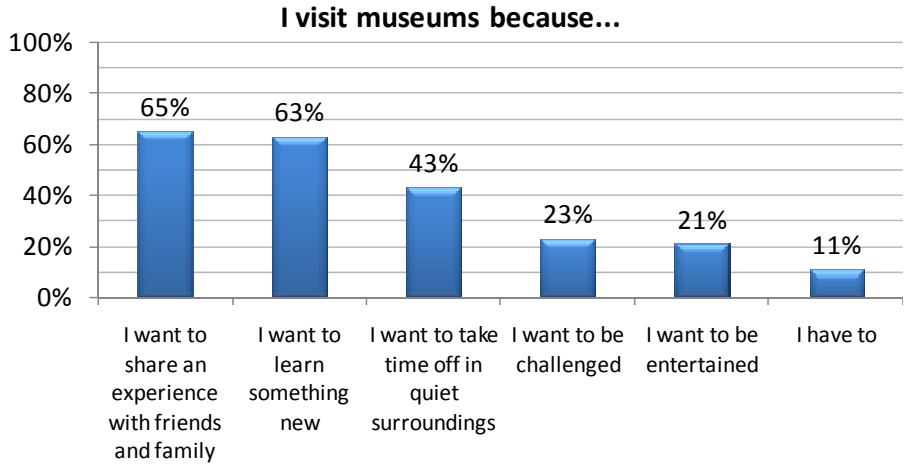


Figure 2: Reasons for visiting museums, N = 138. The respondents could choose three answers.

44 percent of our respondents reported that one of the reasons they do not visit museums is because they do not have the time, implying that these respondents do not consider museum visits as a spare time activity they wish to prioritise. 23 percent answered that they neither find museums and the museum experience interesting nor fun. This could be due to a lack of interactivity and the traditional museum surroundings (Figure 1).

- "I miss interactivity in the exhibitions. More dialogue between the exhibitions and us – all the senses should be stimulated and multiple learning styles available." (male, 34)

Some respondents expressed that "It's a problem you have to read heavy and long texts to get information" (female, 29), and emphasized a feeling of being passive and inactive in museum settings.

We found that the respondents who frequently visit museums find the museum experiences entertaining, thus bringing their family and friends. Still they expect to learn and get challenged, and in this sense strongly indicate that the museums have to facilitate these opportunities (Figure 2).

At NaturBornholm, families with children are the largest visitor group, and the results show that this visitor group stresses the need for more collaborative learning, customisation of information according to age and knowledge level, and social interaction in museums in order to accommodate children's needs:

- "More things children can try, touch and experience." (female, 31)

Despite this, it is important to take into consideration that not all visitors are likely to find the possibilities offered by new technology interesting: “It might be the future, but not for me. I do not think I would know how to use any of it. It is mostly for young people.” (female, 60). Maybe she will never be tempted to use a digital platform, but nevertheless, this emphasizes the need for devices to be easy to use and intuitive, as well as thorough instructions and help on-site.

According to the Danish Ministry of Culture (2008), use of digital media is a necessity to reach the young and potential visitors[6]. Surprisingly, the teenagers in our research did not share this opinion.

- “I do not think technology can make me more interested in museums. Why does everything need to have something to do with technology and mobile phones? It can easily become too much.” (female, 14)

Generally our respondents did not explicitly express a need for digital or technological museum guides and did not comprehend in what way digital technology could contribute to a greater and more engaging experience, thus suggesting that an appealing presentation of the possibilities of such systems is an important prerequisite. In other words, the digital device or technology itself is not sufficient.

[6] http://www.kum.dk/sw75990.asp

2.1 The experience steps

According to our user surveys, a visit to a museum or experience centre usually involves several steps:

1. Before the visit, the visitor may examine the homepage to get specific information and learn about its exhibitions or collections. It is important not only to convey information that the visitor already knows about, but also to provide personalised suggestions on related material that they may not know. Transferring these user preferences seamlessly is crucial to ensure user acceptance.
2. During the visit, the visitors experience the exhibition and may realise new and interesting topics that they would like to explore further. The user-specific preferences should be reflected in the information that is presented to visitors at the experience centre or museum. In terms of guided tours this may result in different data to be presented and even different directions to go.
3. After the stay the visitor may visit locations presented in the museum or experience centre or go home and share the experience with family, colleagues etc.

In developing new interactive guides for museums, experience centres or other cultural heritage institutions these steps should be taken in consideration.

2.2 NaturBornholm

NaturBornholm is an experience centre in Aakirkeby on the Danish island Bornholm. The scope of the centre is to provide easily accessible information about the previous and

present nature of Bornholm to the general public in a dedicated and committed manner. NaturBornholm is both educational and entertaining. The core elements are interactivity, dialogue with nature interpreters, and active participation. To arouse the visitor's curiosity and enhance the experience, all five senses are stimulated.

In the centre's experience hall (Figure 3) the visitors are encouraged to gain hands-on experience in order to develop a greater understanding of the nature on Bornholm. The exhibition hall is a journey through Bornholm's past and present. The exhibition begins with a virtual time travel starting 1 700 millions years ago. The time travel is an animation with additional physical effects, which enable visitors to experience the formation of Bornholm.

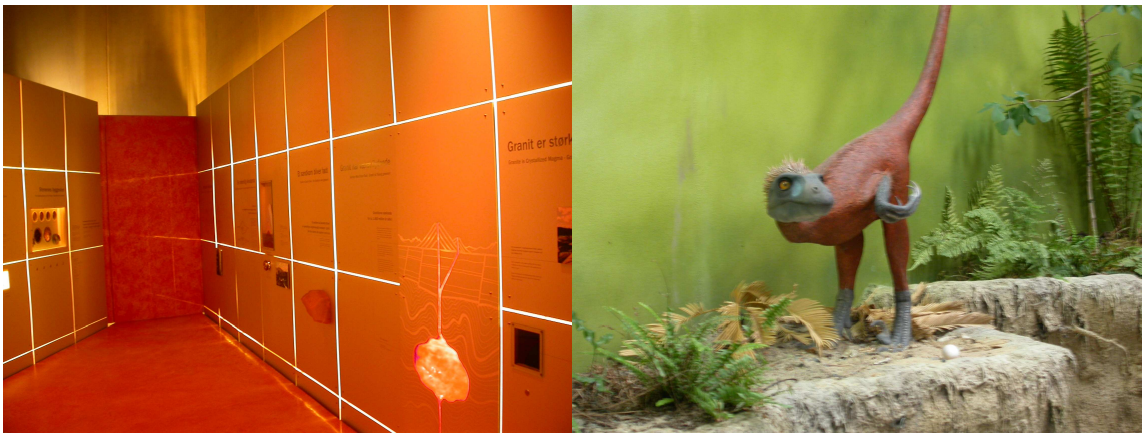


Figure 3: NaturBornholm, indoor.

The largest group of visitors at NaturBornholm is families with children (~96 percent, about 62 000 people annually) who visit the centre as a holiday activity. During

weekdays a great number of schoolchildren and high school students visit the centre as well.

3 Naturl-IT

As mentioned above, the use of handheld devices such as PDAs, multimedia players, and mobile phones as guides within museums is becoming common. At the same time, the telecom statistics from the Danish National IT and Telecom Agency show that approximately 93 percent of the Danish population owns a mobile phone, which is higher than any other portable electronic device (IT- og Telestyrelsen, 2007). Furthermore, mobile phones have a number of relevant features such as camera, GPS, Bluetooth, Web browser, and multimedia player, and the great advantage that users are typically familiar with their device. With this in mind, NaturBornholm chooses to target the mobile phone platform.

Currently NaturBornholm is – similarly to other museums and experience centres – relying on the visitor to bring printed material with them around the museum, and reading it in the vicinity of the intended locations inside the museum or outside. It is therefore a risk that visitors may lose important information by not reading the text in the intended

location, walking in the wrong direction, and become overly distracted or tired of reading large amounts of text (c.f. previous section).

Embedded technologies have great potential to improve the experience both during the visit and afterwards. Mobile devices also have the added benefit of reducing some of the limitations of printed material mentioned above, although they come with a new set of issues such as usability, accessibility and technical issues.

The name of the project presented in this paper is Naturl-IT, and the aim of the project is to offer visitors a possibility to experience and receive information and knowledge about the nature of Bornholm – both inside NaturBornholm and around the island; in particular by displaying audio, video and text at the right location and at the right time taking the user preferences into account. Each visitor has different preferences regarding what he or she likes to see and hear. Among others, there are preferences for the standard school class, family and the advanced visitor interested in geology or botanic. The current implementation relies on GPS for outdoor localization and Bluetooth for indoor localization. The user interface is kept simple with only a few buttons to press on a touch-screen (Figure 4). When a visitor reaches an intended location the phone vibrates for notification. The visitor may therefore keep the mobile phone wherever appropriate and only take it out when needed. The electronic services should not be the prime focus for the visitor, but rather supplement the exhibition and offer an extra dimension.



Figure 4: Screenshot of the Naturl-IT software running on a mobile phone, displaying some multimedia material for a given location.

In order to interact to a greater extent with the visitors, the project also involves the use of 2-dimensional barcodes, more precisely QR-codes ("Quick Response", ISO/IEC 18004:2000). While QR-code (Figure 5) is not the sole 2-dimensional standard, it seems to be the most broadly supported on mobile phones, which are often shipped with built-in readers, for most models in Japan, and some models in Europe, such as Nokia[7] N93, N95, N96 and E90, some telephones based on the Android platform (Google), and other

[7] http://mobilecodes.nokia.com

models depending on mobile phone companies' branding. Furthermore, freeware readers are available for most mobile phone platforms (Java ME, Symbian, Windows Mobile, Google Android, Apple iPhone etc.), e.g. Kaywa[8] Reader, i-nigma[9] Reader and QuickMark[10]. Given some large scale use, in particular for mobile-ticketing, it is reasonable to expect average users to become familiar with this technology in the near future, also in Europe.



Figure 5: A QR-code containing a simple Web address: http://naturbornholm.mobi

[8] http://reader.kaywa.com

[9] http://www.i-nigma.com

[10] http://www.quickmark.com.tw

3.1 Detailed User Scenario

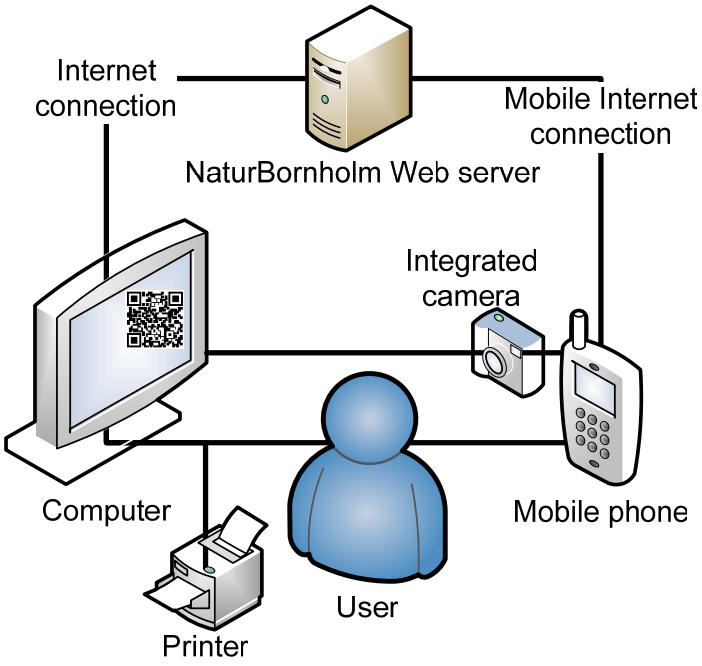


Figure 6: User setup, typically at home preparing the museum excursion.

The suggested user scenario for the Naturl-IT project is divided into three main steps.

1. At home: While preparing the visit the users are offered a set of options to choose from on a Web page. When the selection is done, the preferences are encoded into a QR-code that is displayed on the screen (Figure 6). It is possible to save this tag by printing it on paper, which is particularly useful to distribute the same preferences to a whole group of visitors, e.g. a group of school children.

A more elaborated solution is to transfer those preferences on the user's mobile phone (requires a mobile Internet connection). If the mobile phone is not equipped with a 2D barcode reader, the user is guided to install one. Once the barcode reader is running on

the mobile phone, a simple screenshot of the QR-code displayed on the computer screen will instantly point the mobile phone's Web browser to the same Web page with the same preferences (i.e. Web session), thanks to the barcode containing a specially formatted URL of the following form:

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http://example.net/profile?preferenceA=choice2&preferenceB=choice5
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At this step, the preferences are stored in a cookie on the user's Web browser. If this was not completed from home, or if any modifications are wished, on-site kiosk computers offer the same facilities.

2. At the museum: The Naturl-IT application may be downloaded at no cost from a kiosk. The preferences can be communicated to the kiosk in three manners: by presenting the printed barcode to the kiosk's webcam (not implemented yet), by scanning with the mobile phone a barcode displayed on the kiosk screen (the cookie of step 1 will be received by the kiosk; requires a mobile Internet connection), or by inputting the preferences manually on the kiosk (similar to step 1). The kiosk then broadcasts the application and selected content over a short range Bluetooth connection (10cm) and the user only has to accept it on the phone interface.

3. Outside the museum: When the Naturl-IT application is running, it will automatically react to location changes by providing corresponding information as described in the previous section. Some barcodes containing a location-specific URL may additionally be displayed in selected areas, offering some online multimedia content to users who do not have the Naturl-IT application running (it is at risk of being expensive for users without a suited mobile data plan).

A few technical tricks can be used to make the above procedure more user-friendly. First, it is possible to detect the type of mobile phone (via Bluetooth, and over mobile Web) to provide optimised solutions. If installed, it is also possible to auto-start the Naturl-IT application after the user has scanned a barcode with his/her mobile, by sending to the mobile phone a file with a specific MIME[11] type and extension. Finally, when the Naturl-IT software is running, it can catch some specific Internet addresses to handle them locally (DNS[12] redirection to localhost); this is especially useful to enhance QR-codes that contain a simple URL, in particular to avoid a mobile Internet connection fees.

4 Conclusion

We have presented Naturl-IT; a general principle for location-based services combining Bluetooth, camera and GPS into one system. We have shown that the mobile phone platform can be used for conveying information to visitors both indoor and outdoor. We have used NaturBornholm as a user-case for combining information learned in the exhibitions to those observed outside for preferences-based guided tours. Finally we have presented ideas on how camera-phones can be used for transferring user preferences between mobile phones and computers in a robust and user-friendly manner. It is relevant to note that the authors expect an increase in the proportion of European mobile phones

[11] Multipurpose Internet Mail Extensions

[12] Domain Name System

shipped with a 2D barcode reader pre-installed – as it is already the case in Japan – which will make the proposed procedures easier for the users.

We believe that Naturl-IT is a concept which will bring another dimension to the museum and learning experience at NaturBornholm, and as such, Naturl-IT is a unified whole which is made possible by the technology presented in the paper.

Naturl-IT is not finished yet, and many points are still to be investigated. For instance, location-based games could be a vector to improve user participation and knowledge dissemination. At the moment, the system only supports phones based on Windows Mobile. In time, we hope that other platforms such as Java ME will be supported.

Regarding privacy concerns, it is interesting to note that users carry their own preferences, either or paper or on their mobile phone. There is neither a need for a central database nor for a login with a user account, which reduces privacy threats.

We are working on various ideas for improving the display of data and the interaction principles on the mobile device as this is the major barrier to user acceptance. The use of preferences and providing suggestions to the user will be better implemented, as it is going to play an important role in future museums and experience centres.

5 Acknowledgments

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[13] http://www.danvifo.dk